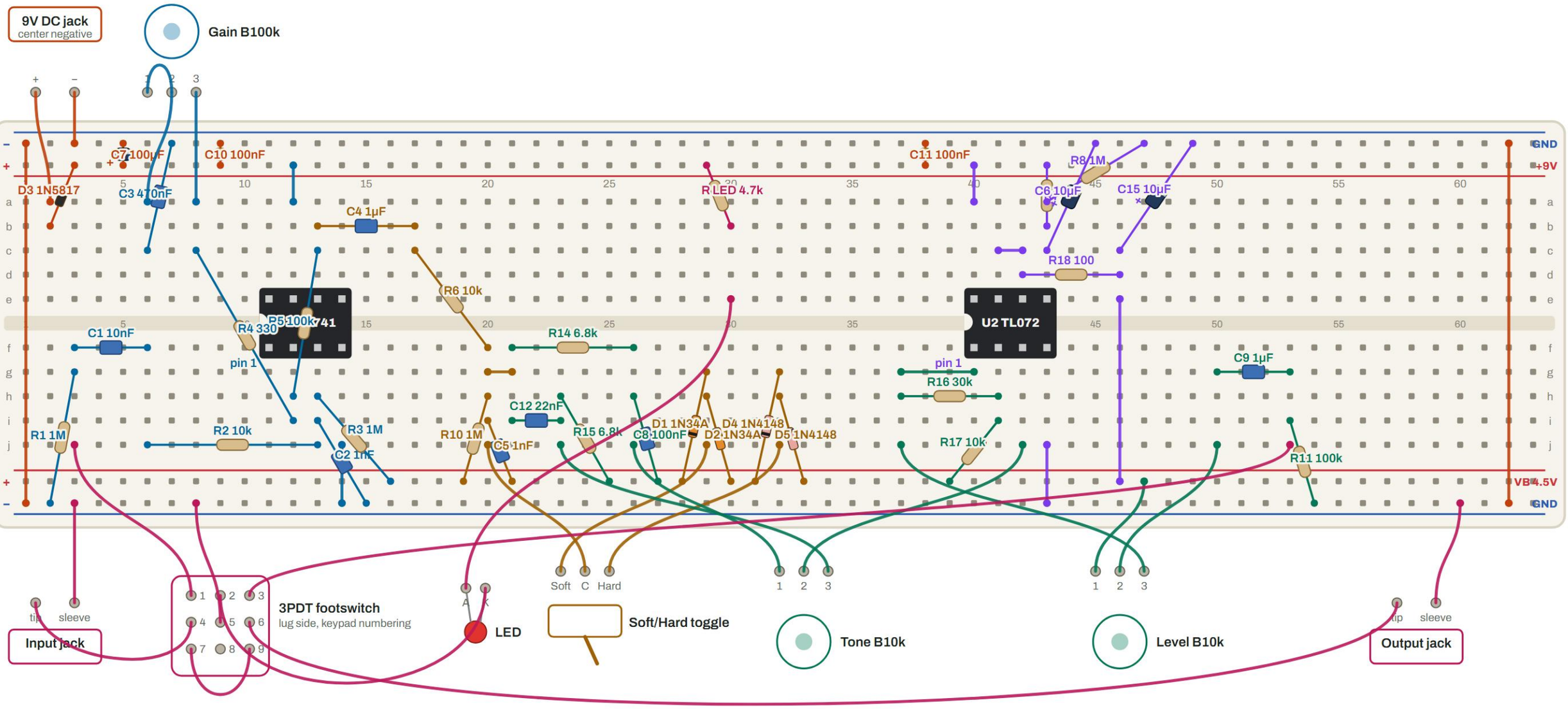
Flashover breadboard, stage 1

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All steps | 1. Rails and power | 2. The 4.5 V bias (VB) | 3. Input and gain stage | 4. Clipping and Soft/Hard | 5. Tone, output amp and Level | 6. Footswitch, LED and jacks



All steps

PART	VALUE	FROM	TO
Step 1	Rails and power	8 items	
Step 2	The 4.5 V bias (VB)	10 items	
Step 3	Input and gain stage	14 items	
Step 4	Clipping and Soft/Hard	12 items	
Step 5	Tone, output amp and Level	15 items	
Step 6	Footswitch, LED and jacks	11 items	

Pick a step above to see its parts, holes and voltage check.

Reading the diagram

- Resistor
- Film capacitor
- Ceramic capacitor
- Electrolytic capacitor (+ marked, pale stripe is –)
- Diode, striped end is the cathode
- Wire or jumper
- Chip, notch on the left, pin 1 at the dot

Hole names: the letter is the row (a–e top half, f–j bottom half), the number is the column. Any hole in the same column and half works the same, so the exact letter only matters for keeping parts from crowding each other.

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Step 1: Rails and power

PART	VALUE	FROM	TO
Jumper	GND link	top blue rail (GND)	bottom blue rail (GND)
Jumper	GND link	top blue rail (GND)	bottom blue rail (GND)
Wire	DC +	DC jack sleeve (+)	a2 (T2)
Wire	DC –	DC jack center (-)	top blue rail (GND)
D3	1N5817 (stripe at To)	b2 (T2)	top red rail (+9V)
C7	100µF (+ at From)	top red rail (+9V)	top blue rail (GND)
C10	100nF	top red rail (+9V)	top blue rail (GND)
C11	100nF	top red rail (+9V)	top blue rail (GND)

> Check: Power up. Black probe on a blue rail: about 8.6–8.8 V on the top red rail, 0 V on the bottom blue rail.

Reading the diagram

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- Film capacitor
- Ceramic capacitor
- Electrolytic capacitor (+ marked, pale stripe is –)
- Diode, striped end is the cathode
- Wire or jumper
- Chip, notch on the left, pin 1 at the dot

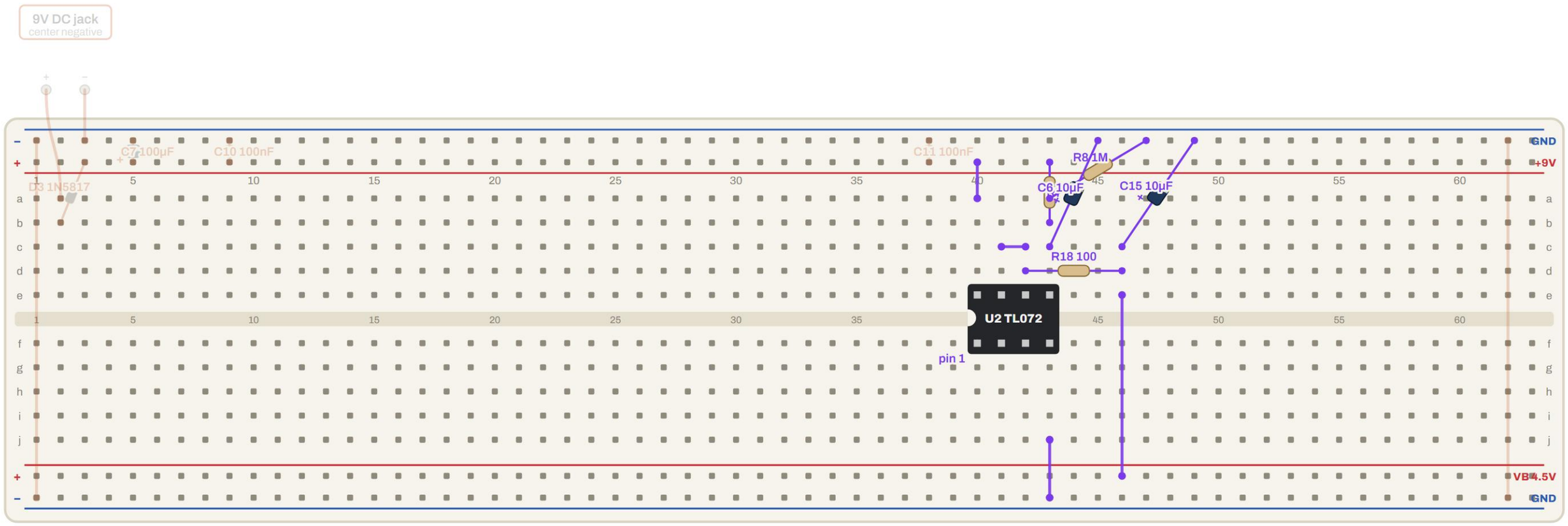
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Flashover breadboard, stage 1

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Step 2: The 4.5 V bias (VB)

PART	VALUE	FROM	TO
U2	TL072	columns 40–43	notch left, pin 1 in f40
Jumper	+9V to pin 8	a40 (T40)	top red rail (+9V)
Jumper	pin 4 to GND	j43 (B43)	bottom blue rail (GND)
R7	1M	b43 (T43)	top red rail (+9V)
R8	1M	a43 (T43)	top blue rail (GND)
C6	10µF (+ at From)	c43 (T43)	top blue rail (GND)
Jumper	pin 7 to pin 6	c41 (T41)	c42 (T42)
R18	100	d42 (T42)	d46 (T46)
C15	10µF (+ at From)	c46 (T46)	top blue rail (GND)
Jumper	VB to rail	e46 (T46)	bottom red rail (VB)

> Check: About 4.5 V on T43, T41 and the bottom red rail (give it a few seconds), and about 8.7 V on T40.

Reading the diagram

- Resistor
- Film capacitor
- Ceramic capacitor
- Electrolytic capacitor (+ marked, pale stripe is –)
- Diode, striped end is the cathode
- Wire or jumper
- Chip, notch on the left, pin 1 at the dot

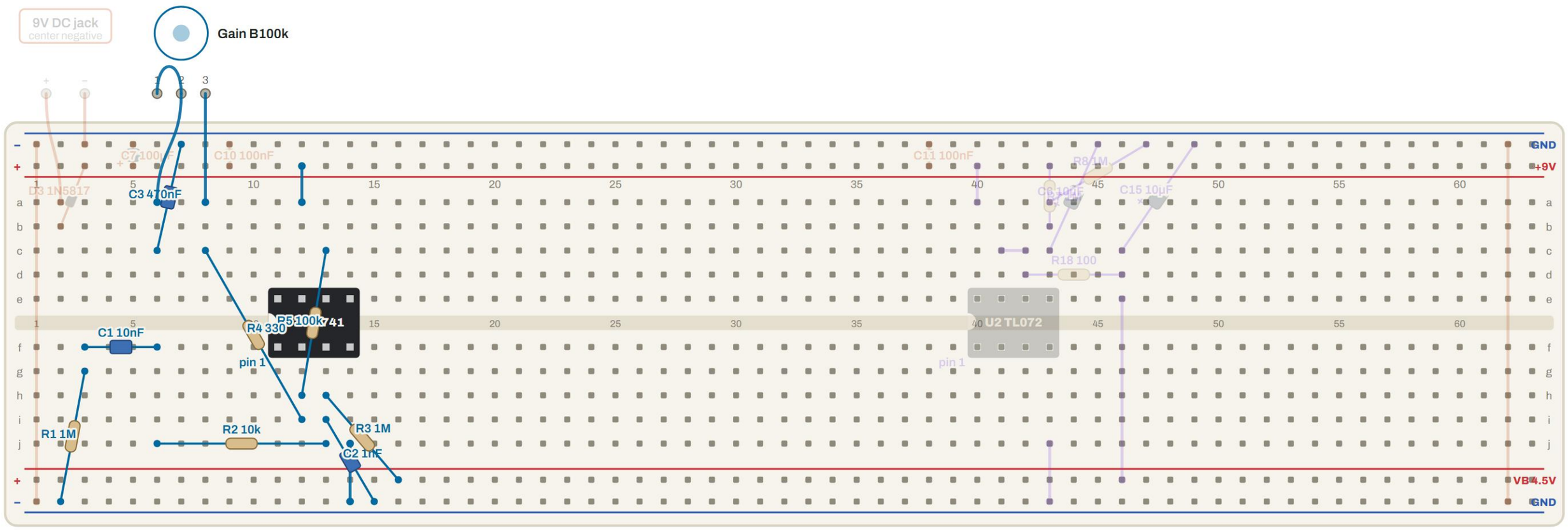
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Flashover breadboard, stage 1

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Step 3: Input and gain stage

PART	VALUE	FROM	TO
U1	LM741	columns 11–14	notch left, pin 1 in f11
Jumper	+9V to pin 7	a12 (T12)	top red rail (+9V)
Jumper	pin 4 to GND	j14 (B14)	bottom blue rail (GND)
R1	1M	g3 (B3)	bottom blue rail (GND)
C1	10nF	f3 (B3)	f6 (B6)
R2	10k	j6 (B6)	j13 (B13)
C2	1nF	i13 (B13)	bottom blue rail (GND)
R3	1M	h13 (B13)	bottom red rail (VB)
R5	100k	h12 (B12)	c13 (T13)
R4	330	i12 (B12)	c8 (T8)
C3	470nF	c6 (T6)	top blue rail (GND)
Wire	Gain lug 3	Gain pot lug 3	a8 (T8)
Wire	Gain lug 2	Gain pot lug 2	a6 (T6)
Link	Gain lug 1 to lug 2	Gain pot lug 1	Gain pot lug 2

> Check: About 4.5 V on T13 (LM741 pin 6) at any Gain setting.

Reading the diagram

- Resistor
- Film capacitor
- Ceramic capacitor
- Electrolytic capacitor (+ marked, pale stripe is –)
- Diode, striped end is the cathode
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- Chip, notch on the left, pin 1 at the dot

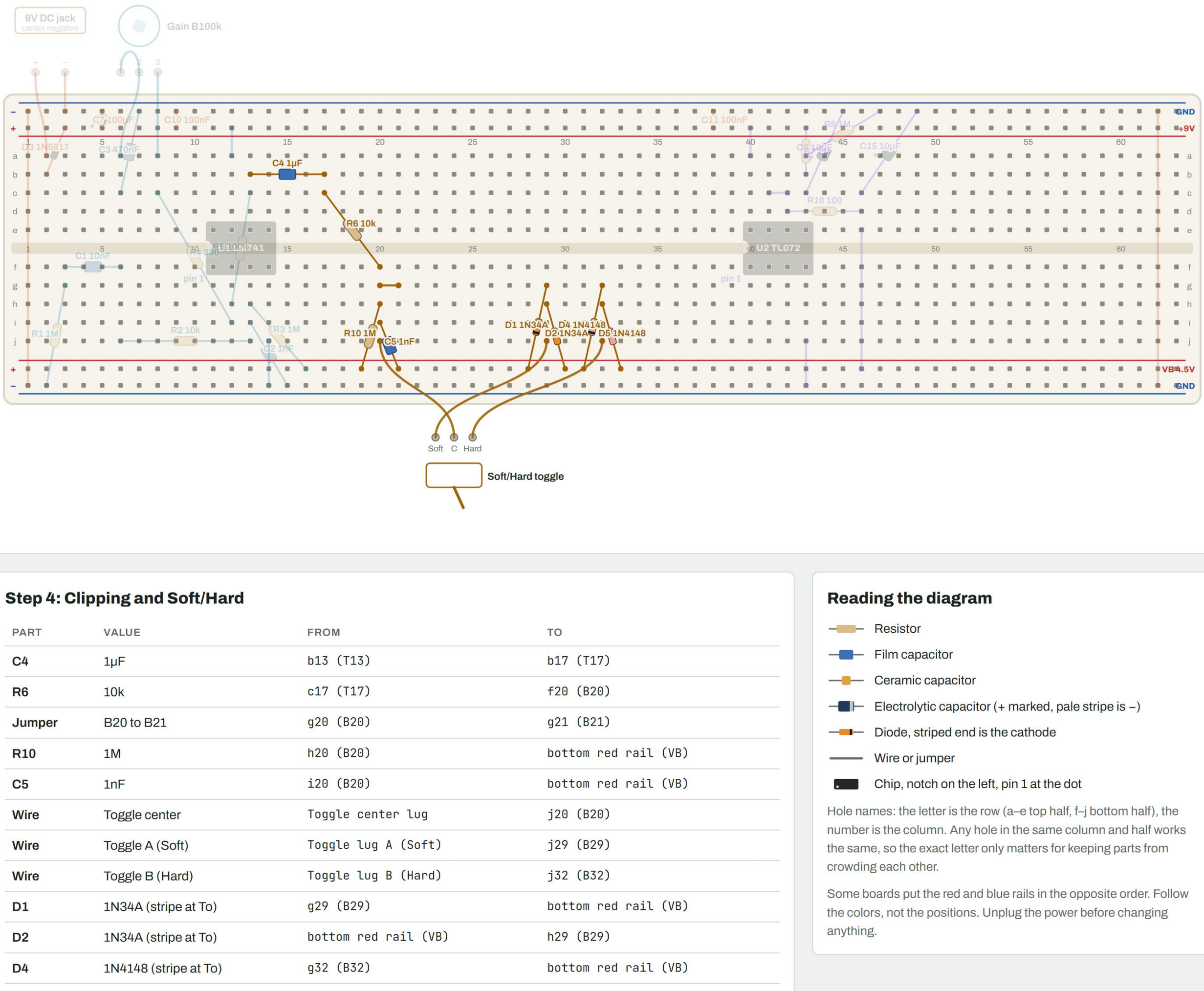
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Step 4: Clipping and Soft/Hard

PART	VALUE	FROM	TO
C4	1µF	`b13 (T13)`	`b17 (T17)`
R6	10k	`c17 (T17)`	`f20 (B20)`
Jumper	B20 to B21	`g20 (B20)`	`g21 (B21)`
R10	1M	`h20 (B20)`	`bottom red rail (VB)`
C5	1nF	`i20 (B20)`	`bottom red rail (VB)`
Wire	Toggle center	`Toggle center lug`	`j20 (B20)`
Wire	Toggle A (Soft)	`Toggle lug A (Soft)`	`j29 (B29)`
Wire	Toggle B (Hard)	`Toggle lug B (Hard)`	`j32 (B32)`
D1	1N34A (stripe at To)	`g29 (B29)`	`bottom red rail (VB)`
D2	1N34A (stripe at To)	`bottom red rail (VB)`	`h29 (B29)`
D4	1N4148 (stripe at To)	`g32 (B32)`	`bottom red rail (VB)`
D5	1N4148 (stripe at To)	`bottom red rail (VB)`	`h32 (B32)`

> Check: About 4.5 V on B20 in both switch positions.

Reading the diagram

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- Film capacitor
- Ceramic capacitor
- Electrolytic capacitor (+ marked, pale stripe is –)
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- Wire or jumper
- Chip, notch on the left, pin 1 at the dot

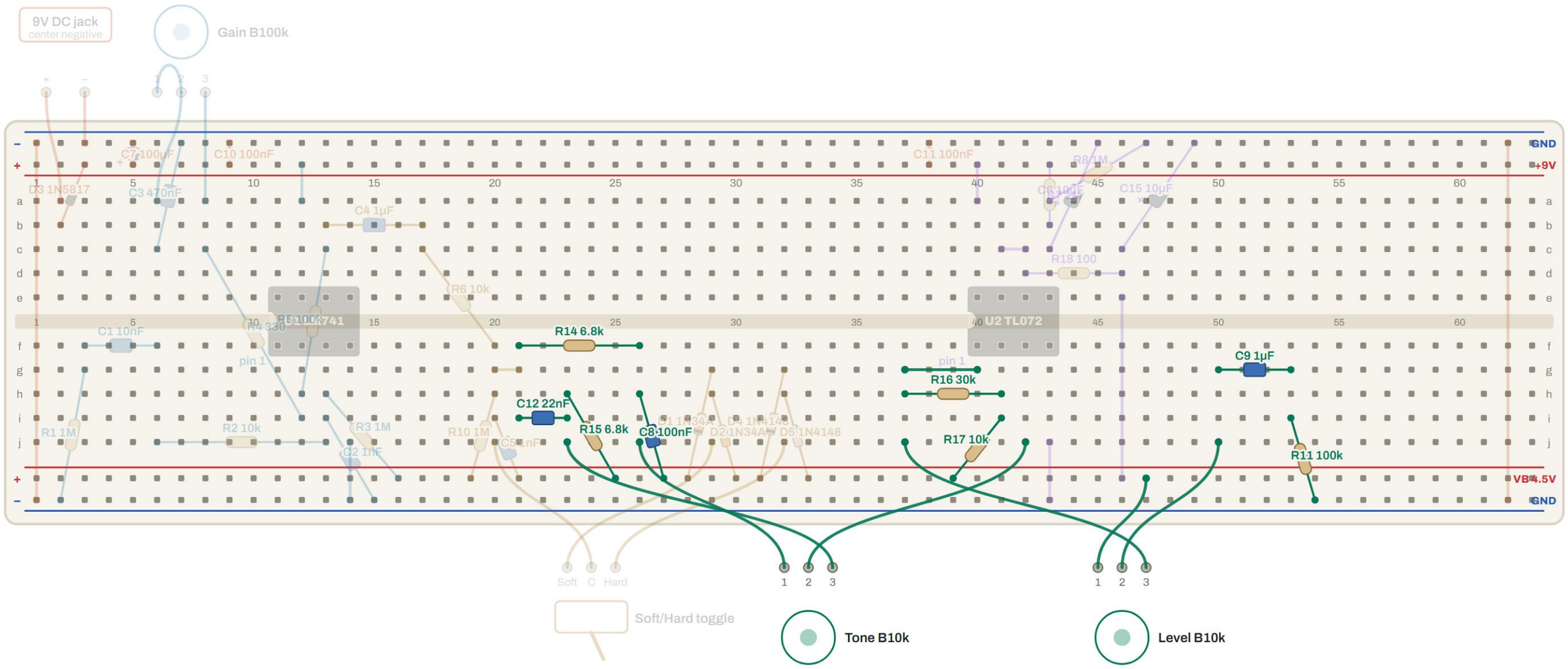
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Step 5: Tone, output amp and Level

PART	VALUE	FROM	TO
C12	22nF	i21 (B21)	i23 (B23)
R15	6.8k	h23 (B23)	bottom red rail (VB)
R14	6.8k	f21 (B21)	f26 (B26)
C8	100nF	h26 (B26)	bottom red rail (VB)
Wire	Tone lug 1	Tone pot lug 1	j26 (B26)
Wire	Tone lug 3	Tone pot lug 3	j23 (B23)
Wire	Tone lug 2	Tone pot lug 2	j42 (B42)
Jumper	B40 to B37	g40 (B40)	g37 (B37)
R16	30k	h37 (B37)	h41 (B41)
R17	10k	i41 (B41)	bottom red rail (VB)
Wire	Level lug 3	Level pot lug 3	j37 (B37)
Wire	Level lug 1	Level pot lug 1	bottom red rail (VB)
Wire	Level lug 2	Level pot lug 2	j50 (B50)
C9	1µF	g50 (B50)	g53 (B53)
R11	100k	i53 (B53)	bottom blue rail (GND)

> Check: About 4.5 V on B40 and B42, and 0 V on B53 (the output).

Reading the diagram

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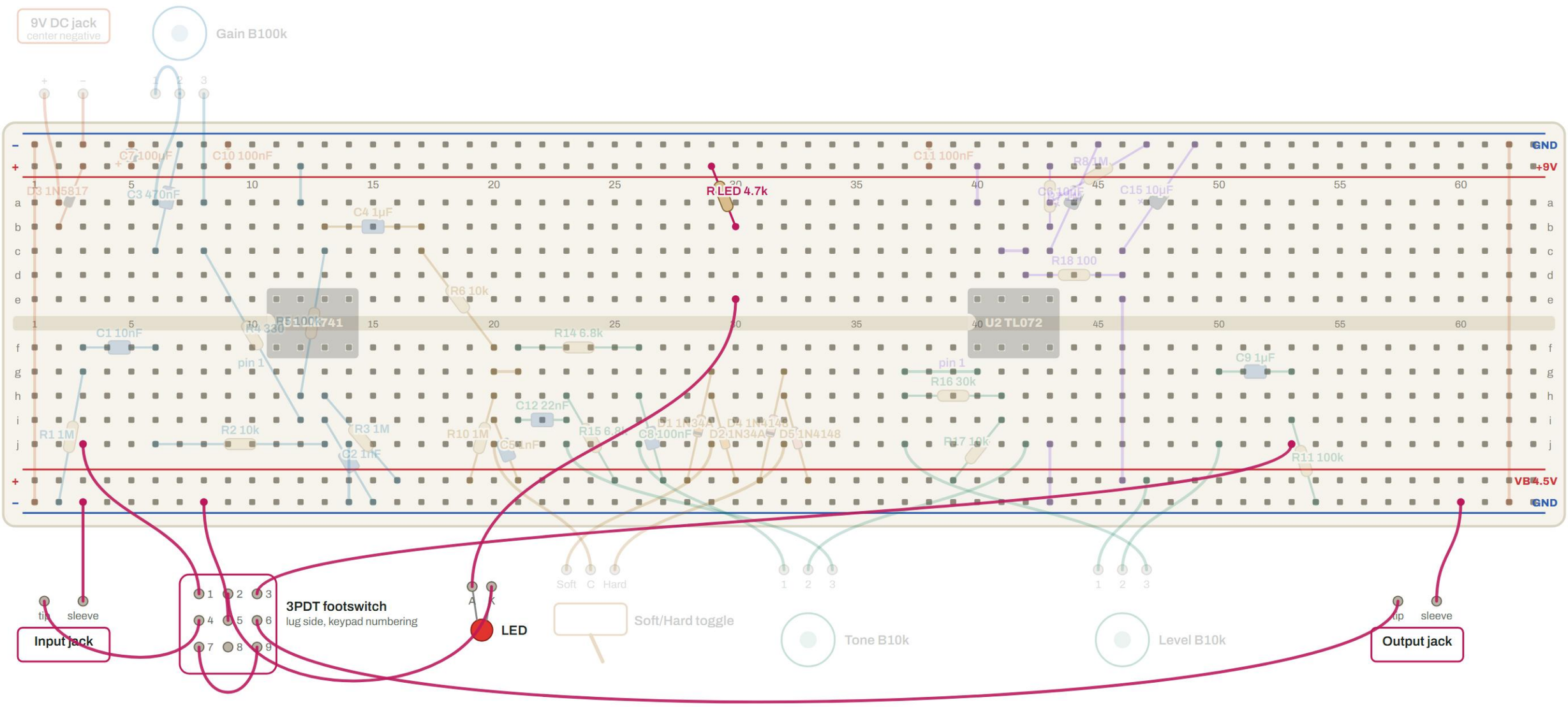
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Step 6: Footswitch, LED and jacks

PART	VALUE	FROM	TO
Link	Lug 7 to lug 9	`Footswitch lug 7`	`Footswitch lug 9`
Wire	Input tip	`Input jack tip`	`Footswitch lug 4`
Wire	Output tip	`Output jack tip`	`Footswitch lug 6`
Wire	LED cathode	`LED short leg (K)`	`Footswitch lug 2`
Wire	Effect in	`Footswitch lug 1`	`j3 (B3)`
Wire	Effect out	`Footswitch lug 3`	`j53 (B53)`
Wire	Footswitch GND	`Footswitch lug 5`	`bottom blue rail (GND)`
R LED	4.7k	`top red rail (+9V)`	`b30 (T30)`
Wire	LED anode	`e30 (T30)`	`LED long leg (A)`
Wire	Input sleeve	`Input jack sleeve`	`bottom blue rail (GND)`
Wire	Output sleeve	`Output jack sleeve`	`bottom blue rail (GND)`

Check: Power off, meter on continuity: the two jack tips beep in one footswitch position (bypass) and not the other. Powered: the LED lights with the effect on.

Reading the diagram

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- Ceramic capacitor
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